

Individual Contribution Report

SurfWatch — AI-Powered Rip Current Detection

31 Mar – 4 Apr 2026

Project	SurfWatch
Student	Alan Bello
Sprint	SCRUM Sprint 1
Period	31 Mar – 4 Apr 2026
Jira Ticket	SCRUM-20 – Create dataset card

1. Introduction

I am a developer on the SurfWatch project, an AI-powered rip current detection system designed to enhance beach safety. This week my primary focus was on completing SCRUM-20: creating a dataset card for the selected rip-current dataset(s) as required by Sprint 1 acceptance criteria.

2. Overview of Contributions

- Researched and identified the RipVIS v1.8.4 dataset (CVPR 2025) as the primary dataset for SurfWatch MVP training and evaluation.
- Created docs/datasets.md in the SurfWatch GitHub repository documenting source, licence, size, annotation format, viewpoint diversity, known issues, and MVP suitability.
- Produced a formatted PDF version of the dataset card for team sharing and Jira submission.
- Reviewed dataset licence (CC BY-NC 4.0) and flagged commercial use restrictions as a risk item for the team.

3. Agile Tools Utilised

Task Boards (Jira)

- Tracked SCRUM-20 on the Jira board throughout the sprint, moving it from To Do → In Progress → Done.
- Added the completed docs/datasets.md path as a comment on SCRUM-20 to satisfy acceptance criteria.

Sprint Planning

- SCRUM-20 was selected during Sprint 1 planning and assigned to me based on my role in data research and documentation.

Version Control (GitHub)

- Uploaded datasets.md to the SurfWatch repo under the docs/ folder, following the structure outlined in the acceptance criteria.

4. Technical Contributions

- Investigated the RipVIS dataset structure: 184 videos (150 with rip currents, 34 without), split into train/val/test by domain experts.
- Documented dual annotation formats supported: COCO JSON and YOLO .txt, with associated train, validation, and test annotation files.
- Identified supplementary training data (2,466 images from CVPRW 2023) bundled within RipVIS and noted its role in the official paper's training pipeline.
- Assessed viewpoint diversity: shore-based cameras, varied weather and sea states, multiple international coastal locations.

- Documented all three known dataset limitations from the official RipVIS README, including the Roboflow annotation jitter issue and frame numbering mismatches.

5. Collaboration and Communication

- Coordinated with the team via Jira to ensure SCRUM-20 was completed within the Sprint 1 deadline.
- Shared the dataset card PDF with team members for review before committing to GitHub.
- Flagged the CC BY-NC 4.0 licence restriction as a team-level concern requiring a decision before commercial deployment.

6. Problem-Solving and Decision Making

- Identified that no docs/ folder existed in the SurfWatch repo and created it as part of this ticket, establishing a documentation structure for future sprints.
- Decided to use the RipVIS CVPR 2025 dataset over alternatives due to its expert annotations, dual-format support, and established evaluation scripts (AP50, F1/F2).
- Proactively included a licence caveat in the MVP suitability section to pre-empt legal risk.

7. Challenges Faced

- The RipVIS dataset has a variable sampling rate and some frame numbering mismatches — required careful reading of the dataset README to accurately document known issues.
- The CC BY-NC 4.0 licence with additional redistribution conditions added complexity; had to carefully interpret what is permissible for SurfWatch's use case.

8. Lessons Learned

- Thoroughly reading a dataset's README before documenting it is essential — the known issues section revealed edge cases that would have been missed otherwise.
- Licence review should be part of the dataset selection process, not an afterthought.
- Creating structured documentation early in the sprint (docs/ folder, consistent file naming) helps the team build good habits for future deliverables.

9. Self-Assessment

- Completed SCRUM-20 in full, meeting all three acceptance criteria: dataset card in repo, all required fields documented, and MVP suitability clearly stated.
- Strength: Thorough and well-structured documentation; went beyond the minimum by including a formatted PDF version.
- Area for growth: Could improve speed of initial research — spent more time than expected verifying dataset licence conditions.

10. Team Feedback

Feedback to be gathered from team members following review of the submitted dataset card. Will update based on any comments received on the Jira ticket or GitHub pull request.

11. Future Goals

- Begin exploring model architecture options for rip current detection using the RipVIS dataset in Sprint 2.
- Resolve the CC BY-NC licence question with the team lead before any training runs are initiated.
- Continue contributing to project documentation to maintain clear traceability between Jira tickets and repository artefacts.

12. Conclusion

This week I successfully completed SCRUM-20 by researching, evaluating, and documenting the RipVIS v1.8.4 dataset as SurfWatch's primary rip current training dataset. The dataset card is now committed to the repository at docs/datasets.md and meets all sprint acceptance criteria. Key findings include the dataset's strong suitability for MVP development, offset by a licence restriction requiring team attention before commercial use. I look forward to building on this foundation in Sprint 2.